

Diễn giải Kết quả Phân tích Dữ liệu

Interpreting Data Analysis Results

Trí tuệ Nhân tạo cho Kế toán (AI in Accounting)

Khoa Kế toán - Kiểm toán

Bài giảng Thực hành 08

Góc nhìn Chuyên gia (Professional Insight)

- **Câu hỏi cốt lõi:** "Liệu kết quả phân tích này có hợp lý không?" (Do the Analysis Results Make Sense?)
- **Trách nhiệm của Kế toán viên:** Kế toán không chỉ tạo ra báo cáo mà còn phải *diễn giải* ý nghĩa của những con số đó để tư vấn cho Ban Lãnh đạo.
- **Tư duy phản biện (Critical Thinking):** Đừng bao giờ tin tưởng tuyệt đối vào kết quả đầu ra của một thuật toán hay AI mà không có sự đánh giá, kiểm chứng từ con người.

Lộ trình Chương (Chapter Roadmap)

- **LO 8.1:** Phân biệt Khám phá và Diễn giải.
- **LO 8.2:** Áp dụng Tư duy Phản biện trong Diễn giải (SPARK-S).
- **LO 8.3:** Đánh giá Sự phù hợp của Kết quả.
- **LO 8.4:** Đánh giá Phân tích Mô tả và Chẩn đoán.
- **LO 8.5:** Đánh giá Phân tích Dự đoán và Đề xuất.

8.1 Rút ra kết luận từ Phân tích dữ liệu

- **Khám phá (Exploration):** Trả lời câu hỏi "Tôi đang nhìn thấy gì?" Mục tiêu là *hiểu dữ liệu* (Understanding the data).
- **Diễn giải (Interpretation):** Trả lời câu hỏi "Điều này có ý nghĩa gì đối với doanh nghiệp?" Mục tiêu là *hiểu kết quả phân tích* (Understanding the analysis).
- **Quá trình Diễn giải:** Đòi hỏi sự kết hợp giữa kiến thức chuyên môn kế toán, hiểu biết về bối cảnh kinh doanh và các nguyên tắc thống kê.

ILLUSTRATION 8.1

Client Accounts Receivable Database Excerpt									
	CustomerID	PaperlessDate	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	Disputed	SettledDate	PaperlessBill
3	0379-NEVHP	4/5/2025	611365	1/1/2025	1/31/2025	\$ 55.94	No	1/14/2025	Paper
4	8976-AMJEO	3/2/2024	7900770	1/25/2025	2/24/2025	\$ 61.74	Yes	3/2/2025	Electronic
5	2820-XGXS	1/24/2024	9231909	7/2/2025	8/1/2025	\$ 65.88	No	7/7/2025	Electronic
6	9322-YCTQO	4/5/2024	9888306	2/9/2025	3/11/2025	\$ 105.92	No	3/16/2025	Electronic
7	6627-ELFBK	11/25/2024	15752855	10/24/2024	11/23/2024	\$ 72.27	Yes	11/27/2024	Paper
8	5148-SYKLB	8/27/2025	18104516	1/25/2024	2/24/2024	\$ 94.00	Yes	2/20/2024	Paper
2466	7050-KQLDO	9/28/2024	9989225541	4/26/2024	5/26/2024	\$ 53.16	No	5/17/2024	Paper
2467	9758-AIEIK	4/22/2024	9990243864	7/3/2025	8/2/2025	\$ 68.66	No	7/17/2025	Electronic
2468									

ILLUSTRATION 8.2

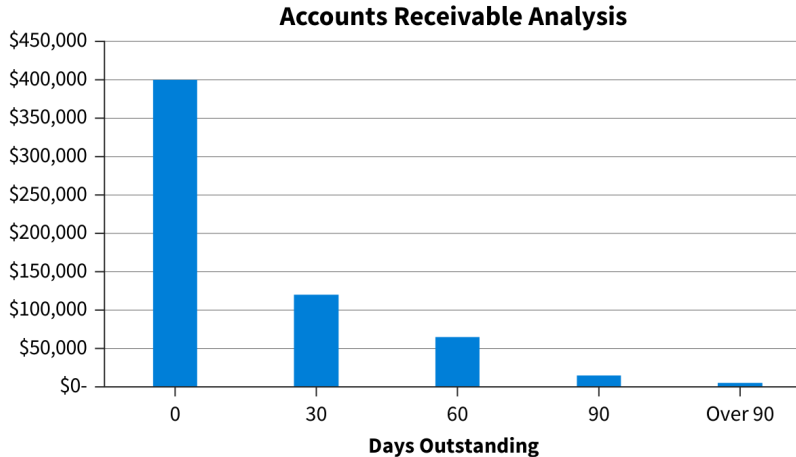


ILLUSTRATION 8.3

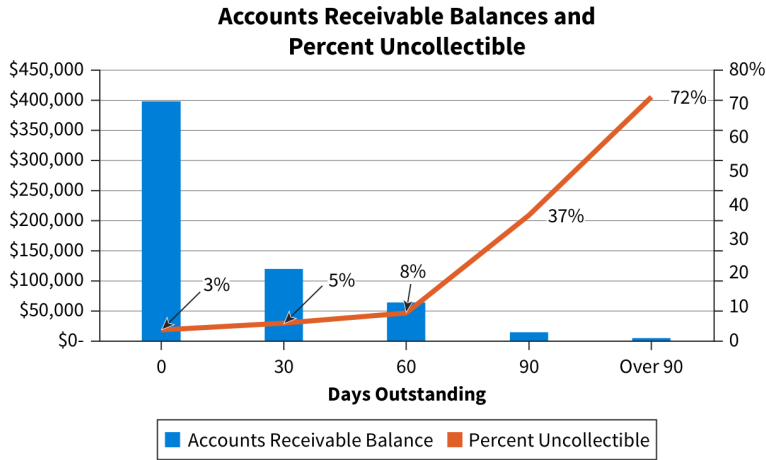


ILLUSTRATION 8.4

Interpretation Process	Example: Interpretation of Illustration 8.23	Learning Objective
Step 1: Determine if the analysis makes sense.		
1. Does the analysis answer the intended question and align with the original objective?	- The question is whether the allowance for doubtful accounts seems reasonable. The analysis does reveal some insight, but more information is needed. - For example, we do not know the balance in the allowance for doubtful accounts.	LO 3
2. Were the correct data and methods used to perform the analysis?	Data: - To make this judgment, check the data to confirm the correct accounts receivable data were used in the analysis. - Confirm that the data used to calculate the uncollectible percentage amounts are correct.	LO 3
	Methods: - The analysis is descriptive in nature. The totals for accounts receivable by due date are provided in the left axis, and the percent uncollectible is illustrated with a call out for each percentage by category. Determine if this is the most appropriate method. - Since the goal is determining if the valuation is reasonable, perhaps a comparison of the allowance for doubtful accounts over the past five years to account receivable balances would be more appropriate.	LO 3
3. Are the results reasonable, or is more analysis necessary?	- To make this judgement, compare the information in the analysis with what is known about the client's accounts receivable. If the totals agree with the financial statements and estimates of uncollectible accounts, then the analysis is reasonable. - We can also evaluate if the visualization is reasonable. For example, most receivables should be in the under 30-day category, and the receivable most likely to not be collectable would be in the 90 days late or over category.	LO 3
Step 2: Verify the results are valid and reliable.		
4. Does the analysis measure what was intended?	Validity refers to how well the analysis represents reality: In this example, determine if the visualization represents an analysis of the allowance for doubtful accounts valuation.	LO 4 and LO 5
5. Are the results accurate?	Accuracy means that the measures used in the analysis are correct and without mistakes: - The measures used in the visualization are account balances, days outstanding, and a calculation of the percent of uncollectible accounts. These are all correct measures. - Confirm the amounts are correct by comparing the accounts receivable to the amounts in the financial statements.	LO 4 and LO 5

Từ Trực quan hóa đến Diễn giải

- **Ví dụ:** Một biểu đồ doanh thu đang đi xuống.
- **Khám phá:** Phát hiện ra doanh thu tháng 10 giảm 15% so với tháng 9.
- **Diễn giải:** Nguyên nhân là do gián đoạn chuỗi cung ứng hoặc thay đổi chính sách tín dụng? Hậu quả là dòng tiền tháng 11 sẽ bị thiếu hụt.

ILLUSTRATION 8.5

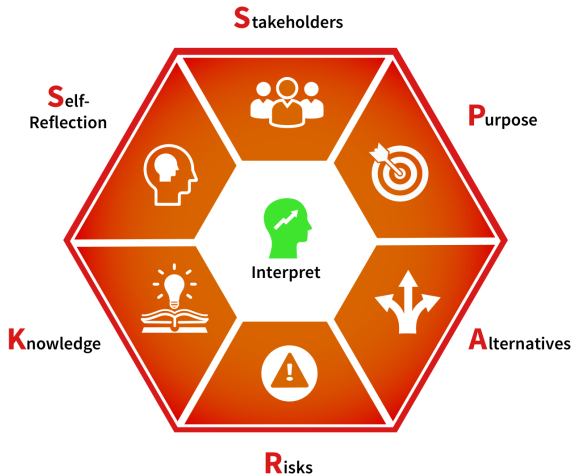


ILLUSTRATION 8.6

Internal Stakeholders

- Super Scooters Managers
- Super Scooters Employees

External Stakeholders

- Investors
- Creditors

ILLUSTRATION 8.7

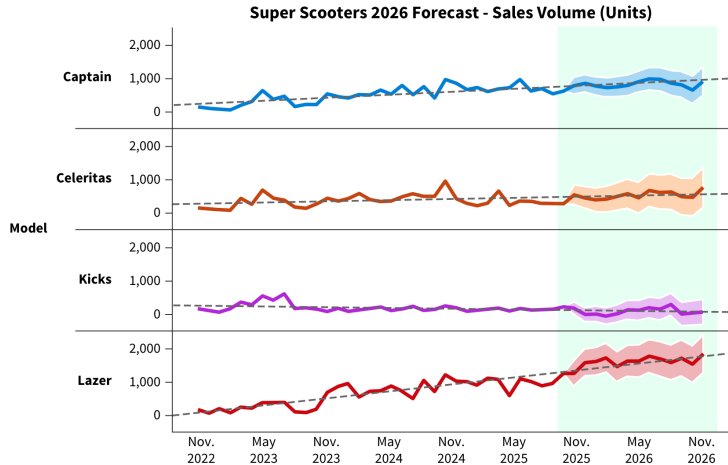


ILLUSTRATION 8.8

Potential Risks	Questions	Super Scooters Examples
Data	• Completeness: Is the analysis missing relevant data? • Accuracy: Are the data used in the analysis correct? • Timeliness: Are the data used in the analysis the most recent available? • Internal controls: Were appropriate internal controls in place to ensure the data used were correct?	• Some of the sales data were not included or may be incorrect. • The data starts in 2022 and continues through 2026, so they appear timely.
Analysis	• Method: Was the correct method used to perform the analysis? • Data: Did the analysis use the right data? • Purpose: Did the analysis answer the question?	• The purpose of the analysis was to understand sales trends for the Super Scooter models. • The analysis uses sales amounts from 2022 to 2025, so it does show how sales are trending. • The risk of using the wrong estimates to forecast the 2026 figures should be considered.
Bias	• Data biases: Were all the necessary and appropriate data included in the analysis? • Preparer biases: Did the preparer have any potential biases that could have influenced the preparation of the analysis? • Evaluator biases: Does the evaluator (you) have any biases that could influence the interpretation of the results?	• The data could be biased if the analysis did not include all the available sales data. • The preparer could be biased to provide a positive analysis if they do not want to eliminate the Celeritas model. • The evaluator could be biased in favor of eliminating the Celeritas or in favor of keeping the Celeritas model.

8.2 Ứng dụng Tư duy Phản biện (SPARK-S)

- Tư duy phản biện giúp chúng ta không bị dẫn dắt bởi những dữ liệu sai lệch hoặc thiên kiến. Khung **SPARK-S**:
 - **S - Stakeholders**: Ai bị ảnh hưởng bởi kết quả này?
 - **P - Purpose**: Mục đích ban đầu của việc phân tích là gì?
 - **A - Alternatives**: Có cách giải thích nào khác cho hiện tượng này không?

ILLUSTRATION 8.9

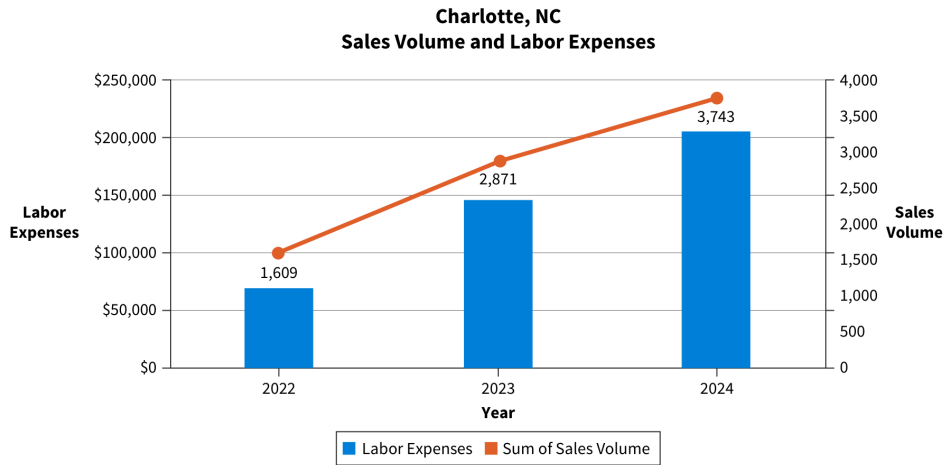


ILLUSTRATION 8.10

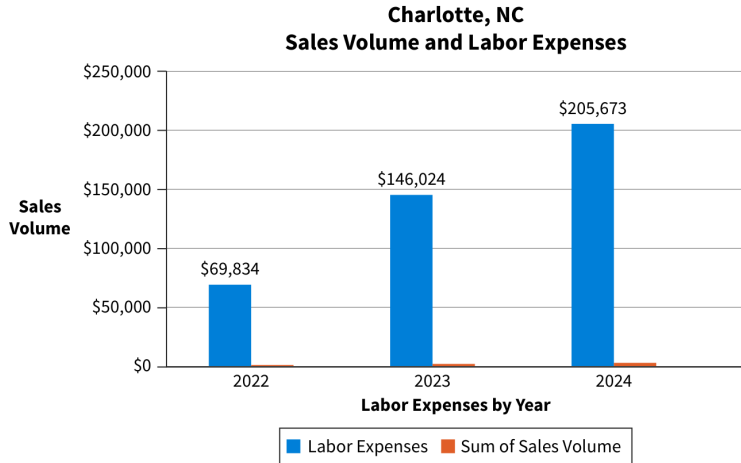


ILLUSTRATION 8.11

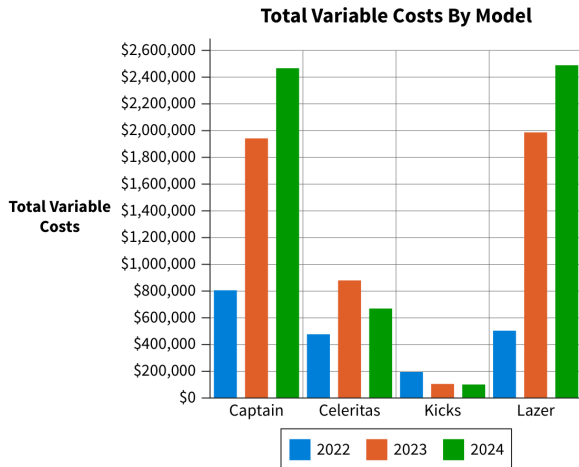
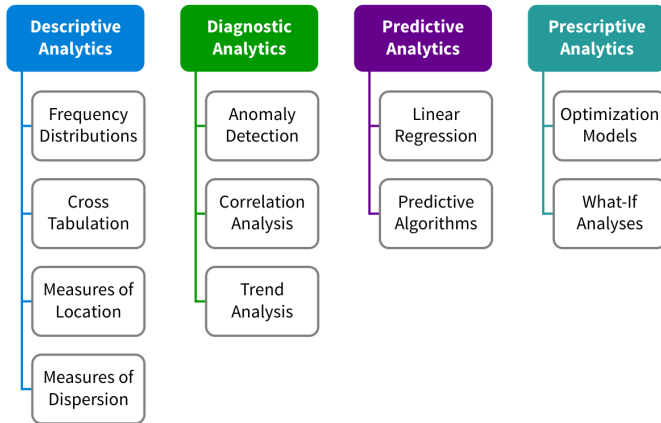


ILLUSTRATION 8.12

Data Analytics Categorized by Purpose



Tư duy Phản biện (Tiếp theo)

- **Khung SPARK-S (Tiếp):**

- **R - Risks & Biases:** Phân tích này có gặp phải thiên kiến xác nhận (Confirmation Bias) hay rủi ro dữ liệu sai không?
- **K - Knowledge:** Chúng ta cần thêm thông tin gì để kết luận chắc chắn hơn?
- **S - Self-reflection:** Đánh giá lại bản thân trong quá trình phân tích.

ILLUSTRATION 8.13

Objective	Example Questions	Valid Analyses
Understand categories of data.	What product category is selling the most? Which business unit has the highest revenue?	Frequency distribution Histogram
Summarize by categories and subcategories.	Which products are selling the most in each region? How much were total expenses each year by business unit?	Cross tabulation analysis
Identify an average observation in the data.	What is the average tax liability for each business unit?	Mean (if no outliers) Median
Evaluate the distribution of data.	How different are the expenses in each business unit from the company average?	Standard deviation

ILLUSTRATION 8.14

Model	Frequency	Relative Frequency
Captain	1,010	27.7%
Celeritas	892	24.5%
Kicks	456	12.5%
Lazer	1,287	35.3%
Total Sales	3,645	100.0%

ILLUSTRATION 8.15

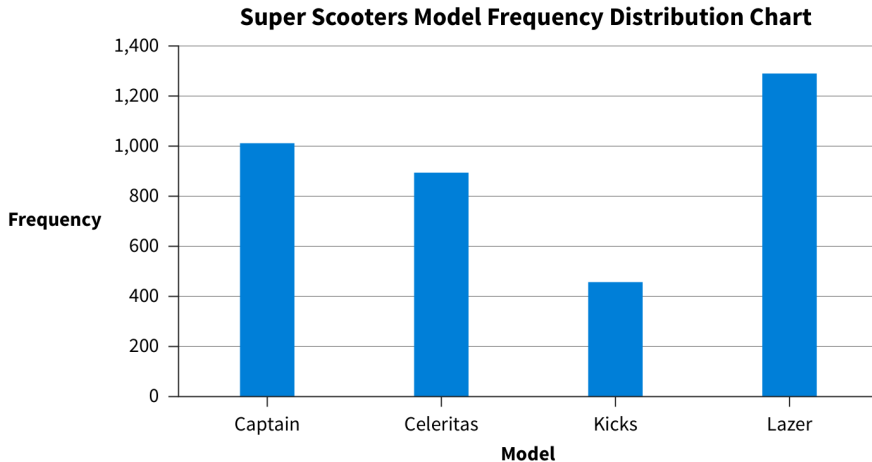


ILLUSTRATION 8.16

Number of Orders Model	Year			Grand Total
	2023	2024	2025	
Captain	244	397	369	1,010
Celeritas	248	327	317	892
Kicks	261	107	88	456
Lazer	207	517	563	1,287
Grand Total	960	1,348	1,337	3,645

8.3 Đánh giá Sự phù hợp của Kết quả

- **Câu hỏi chính:** Kết quả thu được có thực sự *trả lời được câu hỏi kinh doanh* ban đầu không?
- **Đánh giá Dữ liệu (Data):** Dữ liệu có đủ sạch, đủ lớn và đại diện cho vấn đề không?
- **Đánh giá Phương pháp (Methods):** Mô hình thống kê hoặc biểu đồ được chọn có phù hợp với loại dữ liệu không?

ILLUSTRATION 8.17

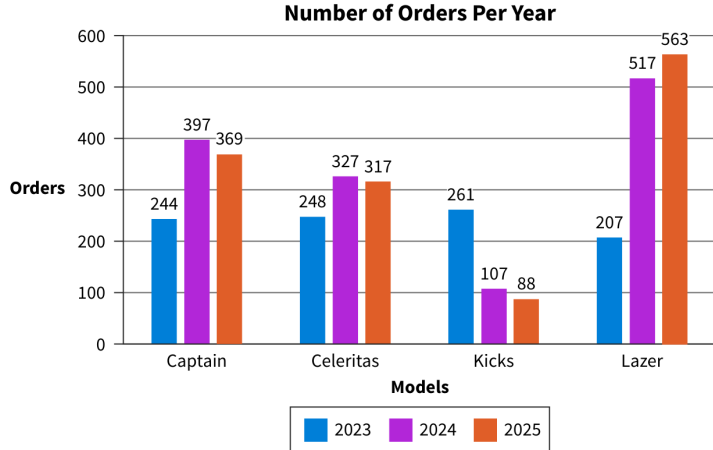


ILLUSTRATION 8.18

Super Scooters, Inc.	
Monthly Sales Volume	
Mean	2,132.39
Standard Error	275.34
Median	1,879.00
Mode	#N/A
Standard Deviation	1,652.02
Sample Variance	2,729,183.96
Kurtosis	8.82
Skewness	2.73
Range	8,050.00
Minimum	371.00
Maximum	8,421.00
Sum	76,766.00
Count	36.00

ILLUSTRATION 8.19

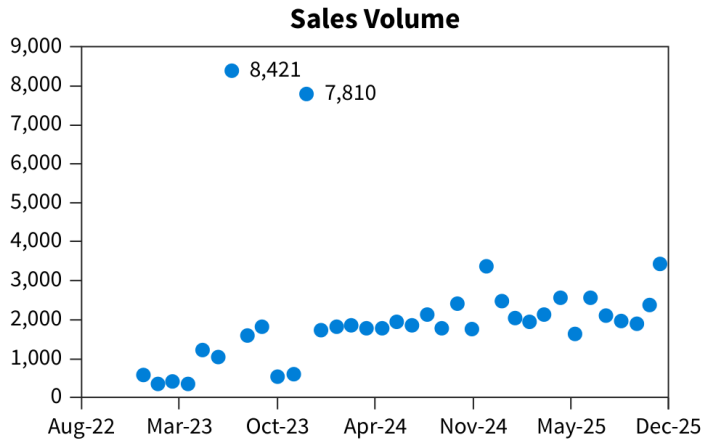


ILLUSTRATION 8.20

Estimated Sales Based on Mean Sales Volume	Estimated Sales Based on Median Sales Volume
$2,132 \times 12 \text{ months} = 25,584$	$1,879 \times 12 \text{ months} = 22,548$
Difference Between Estimated Sales Volume = 3,036	

Kiểm tra và Bổ sung thông tin

- **Kiểm tra Kết quả (Examine Results):** Các con số có ý nghĩa thực tế không? (Ví dụ: Tỷ suất lợi nhuận 500% có thể là do lỗi dữ liệu hơn là thực tế).
- **Xác định thông tin cần thêm:** Có cần bổ sung dữ liệu phi tài chính (Non-financial data) để giải thích cho dữ liệu tài chính không?

ILLUSTRATION 8.21

Super Scooters, Inc.	
Monthly Sales Volume	
Mean	2,132.39
Standard Error	275.34
Median	1,879.00
Mode	#N/A
Standard Deviation	1,652.02
Sample Variance	2,729,183.96
Kurtosis	8.82
Skewness	2.73
Range	8,050.00
Minimum	371.00
Maximum	8,421.00
Sum	76,766.00
Count	36.00

ILLUSTRATION 8.22

Objective	Example Questions	Valid Analyses
Find anomalies in the data set.	Are there revenue transactions that are different or suspiciously high?	Scatterplot Boxplot
Examine relationships in the data.	Is there a relationship between maintenance expense and equipment hours?	Scatterplot Correlation coefficient
Identify patterns in the data set.	Is there a seasonal sales pattern?	Bar chart Column chart Line graph Trend line

ILLUSTRATION 8.23

Correlation r	Interpretation
Exactly -1	A perfect negative linear relationship
-0.70	A strong negative linear relationship
-0.50	A moderate negative linear relationship
-0.30	A weak negative linear relationship
0	No linear relationship
$+0.30$	A weak positive linear relationship
$+0.50$	A moderate positive linear relationship
$+0.70$	A strong positive linear relationship
Exactly $+1$	A perfect positive linear relationship

ILLUSTRATION 8.24

	Correlation Coefficient
Gross Sales	0.9650
Sales Volume	0.7749

8.4 Đánh giá Phân tích Mô tả và Chẩn đoán

- **Tính Hợp lệ (Validity):** Đo lường đúng thứ cần đo.
- **Độ Tin cậy (Reliability):** Tính nhất quán của kết quả khi lặp lại phân tích.
- **Phân tích Mô tả (Descriptive):** Đánh giá tính chính xác của các chỉ số tóm tắt (Tổng, Trung bình, Max, Min).

ILLUSTRATION 8.25

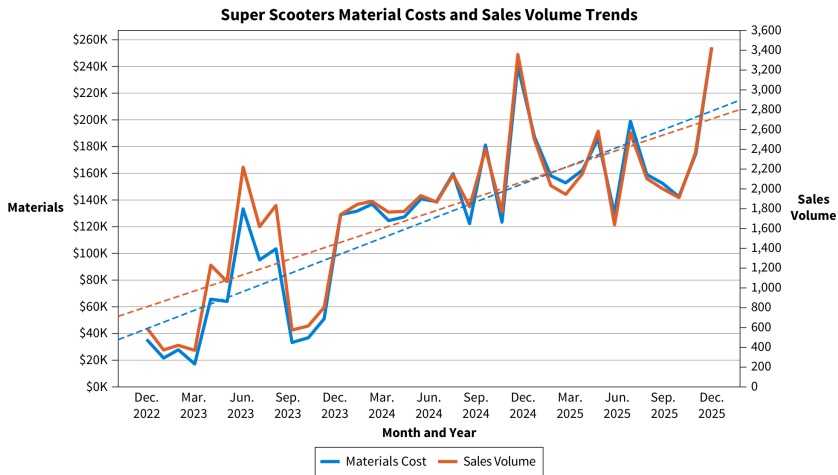


ILLUSTRATION 8.26

Super Scooters Purchasing Department Expense Regression

SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.892897453				
R Square	0.797265861				
Adjusted R Square	0.777957848				
Standard Error	1,337.156824				
Observations	24				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	147659116.6	73829558.29	41.29196782	5.2812E-08
Residual	21	37547755.8	1787988.371		
Total	23	185206872.4			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	-994.5719771	877.1183137	-1.133908575	0.269611889	
# of Purchase Orders	180.0127037	31.32217805	5.747132382	1.05165E-05	
Sales Volume	1.191401172	0.351035522	3.393961854	0.002736568	

ILLUSTRATION 8.27

Regression Statistics	
Multiple R	0.892897453
R Square	0.797265861
Adjusted R Square	0.777957848
Standard Error	1,337.156824
Observations	24

ILLUSTRATION 8.28

Super Scooters Purchasing Department Costs 2022 — 2024

Mean	\$ 3,725.30
Standard Deviation	\$ 2,837.69

Đánh giá Phân tích Chẩn đoán

- **Phân tích Chẩn đoán (Diagnostic):** Giải thích "Tại sao điều đó xảy ra?"
- **Nhận diện Điểm dị biệt (Outliers):** Sử dụng biểu đồ phân tán (Scatterplot) hoặc Box-plot để tìm các giao dịch bất thường trong Kiểm toán.
- **Lưu ý:** Không phải Outlier nào cũng là gian lận, có thể do sai sót nhập liệu.

ILLUSTRATION 8.29

Purchasing Dept. Expenses ANOVA Results

ANOVA					
	df	SS	MS	F	Significance F
Regression	2	147659116.6	73829558.29	41.29197	5.2812E-08
Residual	21	37547755.8	1787988.371		
Total	23	185206872.4			

ILLUSTRATION 8.30

Purchasing Dept. Regression Model

	Coefficients	Standard Error	t Stat	P-value
Intercept	-994.572	877.1183137	-1.133908575	0.269612
# of Purchase Orders	180.0127	31.32217805	5.747132382	1.05E-05
Sales Volume	1.1914012	0.351035522	3.393961854	0.002737

ILLUSTRATION 8.31

Purchasing Dept. Predicted Expenses

	Model Coefficient	Variable Values	Prediction
Intercept	\$ (994.57)	1	\$ (994.57)
# of Purchase Orders	\$ 180.01	12	\$ 2,160.15
Sales Volume	\$ 1.19	2,200	\$ 2,621.08
			\$ 3,786.66

ILLUSTRATION 8.32

	A	B	C
1	Ballet Nuevo-Additional Performance Scenario Analysis		
2			
3	Expected Ticket Sales	2,500	people
4	Average Concessions Spending	\$ 25.00	per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 50.00	per person
7	Concession Profit %	50%	
8	Profit Percentage	80%	
9			
10	Ticket Revenue	\$ 125,000.00	

Theater charges, based on ticket sales, are expenses charged to the ballet company from the theater. The higher the ticket sales, the lower the theater charge

Expected ticket sales and concession spending (food and drink) by ticket holders

The theater takes 20% of the ticket price and 50% of concession sales

8.5 Đánh giá Phân tích Dự đoán và Đề xuất

- **Phân tích Dự đoán (Predictive):** Sử dụng mô hình (như Hồi quy - Regression) để dự báo tương lai.
- **Đánh giá Mô hình:**
 - Hệ số R^2 có đủ cao không?
 - Mỗi quan hệ có ý nghĩa thống kê ($p\text{-value} < 0.05$) không?
 - Cảnh thận với "Tương quan không có nghĩa là Nhân quả" (Correlation vs. Causation).

ILLUSTRATION 8.32_1

AutoSave OFF

	A	B	C
1	Ballet Nuevo-Additional Performance Scenario Analysis		
2			
3	Expected Ticket Sales	2,500	people
4	Average Concessions Spending	\$ 25.00	per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 50.00	per person
7	Concession Profit %	50%	
8	Profit Percentage	80%	
9			
10	Ticket Revenue	\$ 125,000.00	
11	Concession Revenue	\$ 31,250.00	
12			
13	Profit	\$ 113,250.00	

Sheet1

Theater charges, based on ticket sales, are expenses charged to the ballet company from the theater. The higher the ticket sales, the lower the theater charge

Expected ticket sales and concession spending (food and drink) by ticket holders

The theater takes 20% of the ticket price and 50% of concession sales

Ballet Nuevo Scenario Analysis Model

	Likely	Optimistic	Pessimistic
Expected Ticket Sales	2,500	4,500	1,500
Concessions Spending	\$ 25.00	\$ 40.00	\$ 10.00
Theater Charges	\$ 18,000.00	\$ 10,000.00	\$ 25,000.00

ILLUSTRATION 8.34

Ballet Nuevo Scenario Analysis Results

Scenario Summary				
	Current Values	Optimistic	Pessimistic	Likely
Changing Cells				
\$B\$3	2,500	4,500	1,500	2,500
\$B\$4	\$ 25.00	\$ 40.00	\$ 10.00	\$ 25.00
\$B\$5	\$ 18,000.00	\$ 10,000.00	\$ 25,000.00	\$ 18,000.00
Result Cells				
\$B\$13	\$ 113,250.00	\$ 260,000.00	\$ 42,500.00	\$ 113,250.00

ILLUSTRATION 8.35

The screenshot shows an Excel spreadsheet with a 'Goal Seek' dialog box open. The spreadsheet data is as follows:

	A	B	C
1	Ballet Nuevo – Profit Target Goal Seek		
2			
3	Expected Ticket Sales	2,500	People
4	Average Concessions Spending	\$ 25.00	Per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 50.00	Per person
7	Concession Profit %	50%	
8	Profit percentage	80%	
9			
10	Ticket Revenue	\$ 125,000.00	
11	Concession Revenue	\$ 31,250.00	
12			
13	Profit	\$ 113,250.00	

The 'Goal Seek' dialog box is configured with the following values:

- Set cell:** \$B\$13
- To value:** 150000
- By changing cell:** \$B\$6

The 'OK' button is highlighted.

ILLUSTRATION 8.36

The screenshot shows an Excel spreadsheet with a green title bar. The spreadsheet has three columns: A, B, and C. The data is as follows:

	A	B	C
1	Ballet Nuevo – Profit Target Goal Seek		
2			
3	Expected Ticket Sales	2,500	People
4	Average Concessions Spending	\$ 25.00	Per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 68.38	Per person
7	Concession Profit %	50%	
8	Profit percentage	80%	
9			
10	Ticket Revenue	\$ 170,937.50	
11	Concession Revenue	\$ 31,250.00	
12			
13	Profit	\$ 150,000.00	

The spreadsheet interface includes a formula bar at the top with 'fx' and a status bar at the bottom showing 'Sheet1'.

ILLUSTRATION 8.37

Be sure you check the box before "Sales Transactions 2024-2025." If you forget you will not have the option to "Load."

Navigator

Display Options

- SuperScootersSales_2024-2025.xlsx [2]
- Sales Transactions 2024-2025
- Sheet2

Sales Transactions 2024-2025

Order Number	Year	Sales Order Number	Model	Sold Date
13684	2024	13684-2024	Celeritas	12/31/2024
13685	2024	13685-2024	Lazer	12/31/2024
13682	2024	13682-2024	Captain	12/30/2024
13683	2024	13683-2024	Lazer	12/30/2024
13677	2024	13677-2024	Lazer	12/29/2024
13678	2024	13678-2024	Lazer	12/29/2024
13679	2024	13679-2024	Celeritas	12/29/2024
13680	2024	13680-2024	Captain	12/29/2024
13681	2024	13681-2024	Celeritas	12/29/2024
13675	2024	13675-2024	Celeritas	12/28/2024
13676	2024	13676-2024	Captain	12/28/2024
13671	2024	13671-2024	Captain	12/27/2024
13672	2024	13672-2024	Lazer	12/27/2024
13673	2024	13673-2024	Lazer	12/27/2024
13674	2024	13674-2024	Captain	12/27/2024
13668	2024	13668-2024	Captain	12/26/2024
13669	2024	13669-2024	Celeritas	12/26/2024
13670	2024	13670-2024	Kicks	12/26/2024
13665	2024	13665-2024	Lazer	12/25/2024

The data in the preview has been truncated due to size limits

Load Transform Data Cancel

Đánh giá Phân tích Đề xuất

- **Phân tích Đề xuất (Prescriptive):** Tối ưu hóa các nguồn lực để đạt kết quả tốt nhất.
- **Đánh giá Khả năng Thực thi:** Đề xuất của AI có khả thi trong điều kiện ngân sách và chính sách của công ty không?
- **Kết luận:** Con người luôn là chốt chặn cuối cùng trong quá trình ra quyết định tài chính.

ILLUSTRATION 8.38

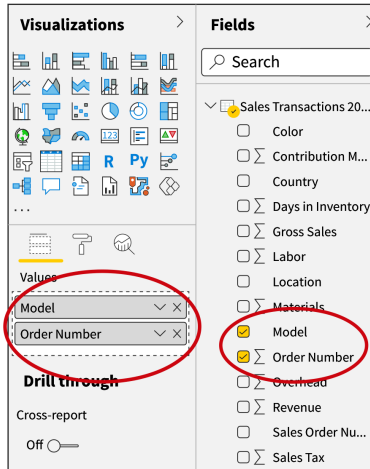


ILLUSTRATION 8.39

Model	Frequency	Relative Frequency
Captain	1,010	27.71%
Celeritas	892	24.47%
Kicks	456	12.51%
Lazer	1,287	35.31%
Total Sales	3,645	100.00%

ILLUSTRATION 8.40

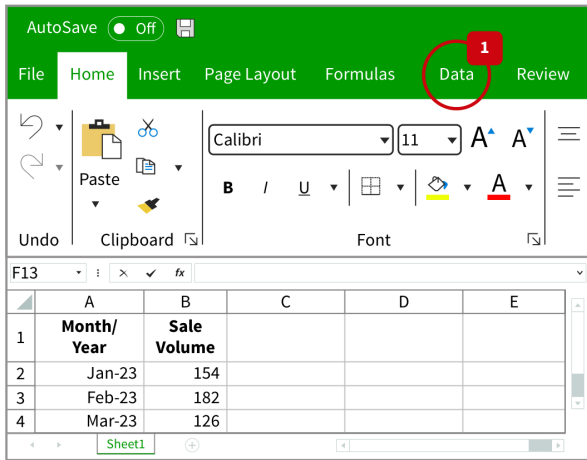


ILLUSTRATION 8.41

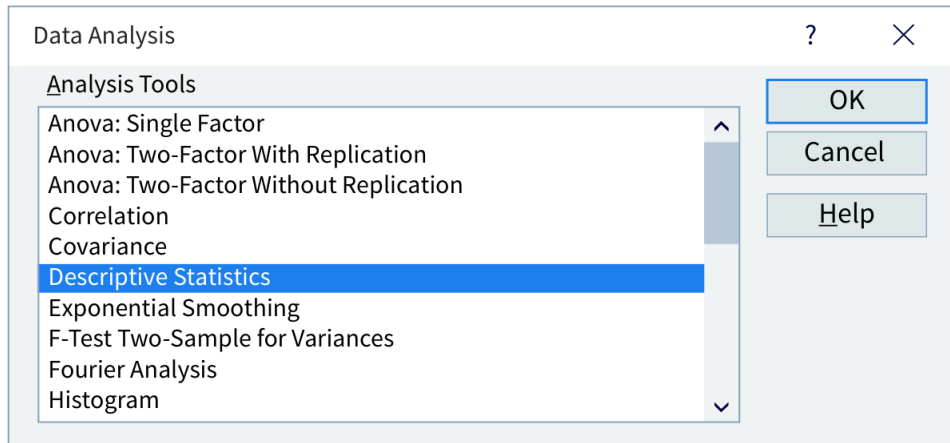


ILLUSTRATION 8.42

Descriptive Statistics

Input

Input Range: **3** \$B\$1:\$B\$37

Grouped By: ☒ Columns ☐ Rows

☒ Labels in First Row

Output options

☐ Output Range: **4**

☒ New Worksheet Ply:

☐ New Workbook

☒ Summary statistics

☐ Confidence Level for Mean: 95 %

☐ Kth Largest: 1

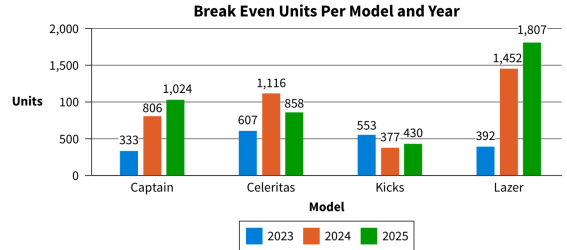
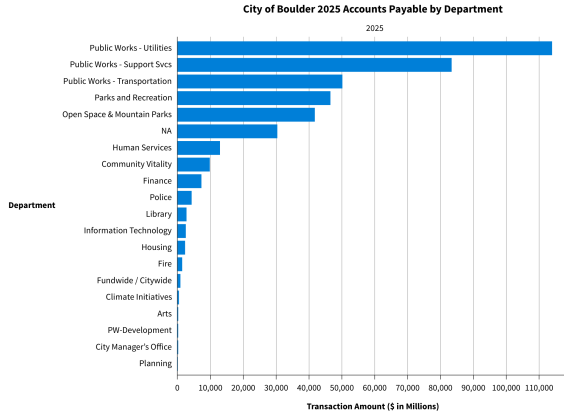
☐ Kth Smallest: 1

OK Cancel Help

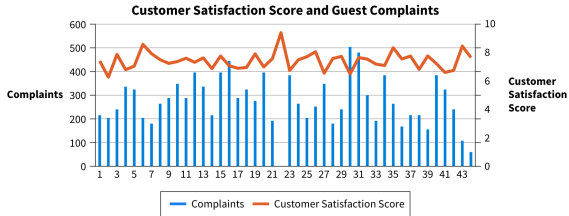
Phần Bài tập Ngắn

Brief Exercises (BE)

BE 8.3 & BE 8.4



BE 8.7 & BE 8.8



U.S. Outdoor Adventure Stores	
Sales 2022–2025	
Mean	229.86
Median	54.49
Mode	12.96
Standard Deviation	623.25
Range	22,638.04
Minimum	0.44
Maximum	22,638.48
Count	9,994.00

BE 8.9 & BE 8.10

	Correlation: Customer Satisfaction Score
Customer Satisfaction Score	1
Annual Customer Complaints	-0.826124807
Minutes Spent at Property	0.938922087
Number of Employees	0.681057818

Summary Output	
Regression Statistics	
Multiple R	0.87254918
R Square	0.76134208
Adjusted R Square	0.76098337
Standard Error	158,773.661
Observations	2,000

Net Revenue

	2025
Gross Sales	\$ 12,165,162
Total Variable Costs	\$ 5,722,777
Contribution Margin	\$ 6,442,385
Total Fixed Costs	\$ 1,696,822
Net Revenue	\$ 4,745,563

What-If Analysis

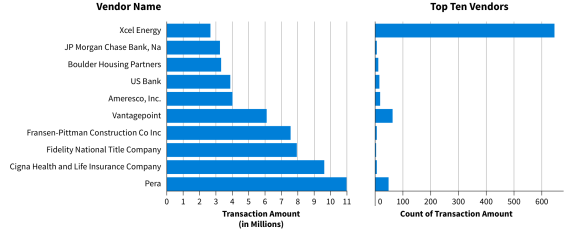
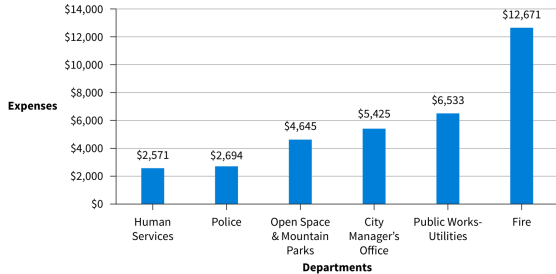
	Best-Case	Likely-Case	Worst-Case
Gross Sales	\$ 12,165,162	\$ 12,165,162	\$ 12,165,162
Total Variable Costs	\$ 6,008,916	\$ 5,837,232	\$ 6,123,371
Contribution Margin	\$ 6,156,246	\$ 6,327,930	\$ 6,041,791
Total Fixed Costs	\$ 1,696,822	\$ 1,696,822	\$ 1,696,822
Net Revenue	\$ 4,459,424	\$ 4,631,108	\$ 4,344,969

Phần Bài tập

Exercises (EX)

EX 8.1 & EX 8.2

Business Expenses - Top Six Departments

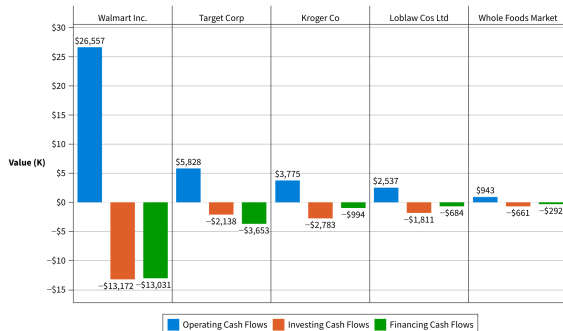


EX 8.4 & EX 8.5

Super Scooters, Inc.
Accounts Receivable Aging Table

Customer	Value				
C1036	\$40K OK	\$7,184	0	0	0
C1282	\$40K OK	0	0	0	\$32,478
C1423	\$40K OK	\$17,635	0	0	0
C1671	\$40K OK	\$23,038	0	0	0
C2036	\$40K OK	\$763	0	0	0
C2274	\$40K OK	\$23,562	0	0	0
C3168	\$40K OK	0	0	0	\$32,229
C9591	\$40K OK	\$10,840	0	0	0
C9917	\$40K OK	0	0	0	\$31,046
		Less than 30	31 to 60	61 to 90	91 to 180
		Days			
		More than 180...			

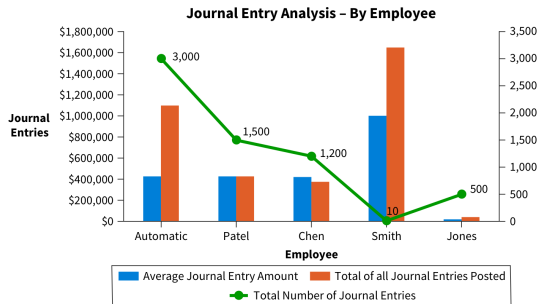
Cash Flow



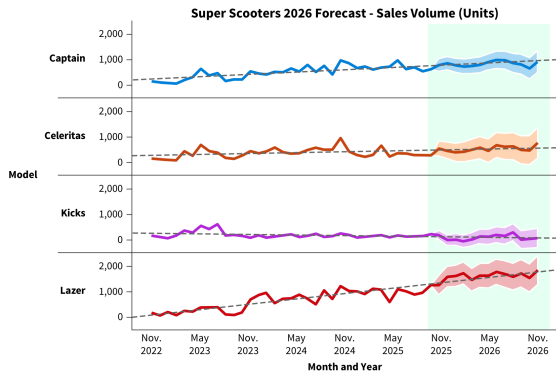
EX 8.6 & EX 8.7

Profit Margin by Customer				
Cust Name	2021	2022	2023	2024
Bluebird Cafe	45.83%	45.15%	47.28%	53.04%
Butters AM Eatery	44.85%	21.98%	45.51%	50.90%
Home Cookin Café	42.77%	39.14%	48.17%	55.14%
Krispy Kreme	48.84%	49.21%	31.18%	47.92%
Lucile's Creole Café	46.52%	48.44%	49.44%	40.91%
Red Rooster Restau..	44.31%	41.04%	50.38%	50.02%
Rocky Mountain Bag..	50.02%	45.32%	47.82%	40.53%
Snooze AM Eatery	47.11%	47.41%	47.37%	33.27%
Syrup Downtown	42.23%	56.82%	43.61%	43.47%
Ziggi's Coffee	41.69%	42.60%	56.57%	38.71%

Avg. Profit Margin



EX 8.8 & EX 8.9



Model	Sum of Gross Sales
Captain	\$ 12,324,133
Celeritas	\$ 4,792,338
Kicks	\$ 1,108,183
Lazer	\$ 8,922,753
Grand Total	\$ 27,147,407

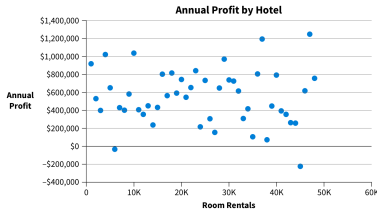
EX 8.10 & EX 8.11

Yearly Region Sales				
Region	2022	2023	2024	2025
Central	\$103,838	\$102,874	\$147,429	\$147,098
East	\$128,680	\$156,332	\$180,529	\$213,239
South	\$103,846	\$ 71,301	\$ 93,539	\$122,977
West	\$147,883	\$139,766	\$186,976	\$250,633



Cost Drivers	Beds	Outpatient Visits	Births	Gen Med / Surg or Not	Total Expense
Beds	1				
Outpatient Visits	0.61584604	1			
Births	0.66665244	0.496333644	1		
Gen Med / Surg or Not	0.25764602	0.322183655	0.359453	1	
Total Expense	0.77739236	0.782692108	0.62695	0.249571064	1

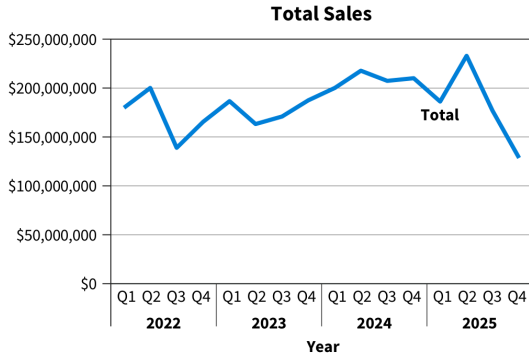
EX 8.12 & EX 8.13



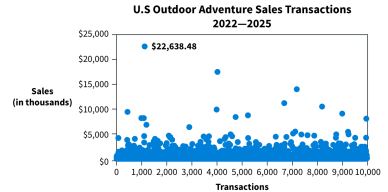
Profit	
Mean	547,691.05
Standard Error	44,607.48
Median	553,868.00
Mode	#N/A
Standard Deviation	309,049.66
Sample Variance	95,511,689,341.06
Kurtosis	0.07
Skewness	0.00
Range	1,472,109.00
Minimum	(224,305.00)
Maximum	1,247,804.00
Sum	26,289,170.25
Count	48.00

Row Labels	Sum of Total Sales
Canada	\$ 788,914,146
Offline	\$ 392,731,724
Online	\$ 396,182,421
Mexico	\$ 710,979,369
Offline	\$ 400,360,732
Online	\$ 310,618,637
United States	\$ 1,452,693,443
Offline	\$ 720,489,036
Online	\$ 732,204,407
Grand Total	\$ 2,952,586,958

EX 8.14 & EX 8.15



Descriptive Statistics		
Sales 2022 — 2025		
Mean		229.86
Median		54.49
Mode		12.96
Standard Deviation		623.25
Sample Variance		388,434.46
Minimum		0.44
Maximum		22,638.48
Count		9,994.00



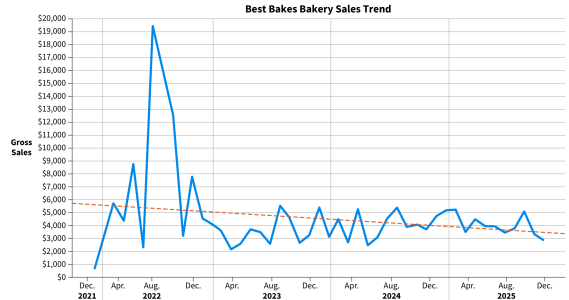
EX 8.16 & EX 8.17A

Hospitality, Inc - Total Expenses Prediction Model					
SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.810435721				
R Square	0.656806059				
Adjusted R Square	0.62160668				
Standard Error	79,448.54158				
Observations	44				
ANOVA					
	df	SS	MS	F	Significance F
Regression	4	4.71E+11	1.18E+11	18.65959	1.211E-08
Residual	39	2.46E+11	6.31E+09		
Total	43	7.17E+11			
	Coefficients	Standard Error	t Stat	P-value	
Intercept	(131,136.43)	121071.9	-1.08313	0.285403	
Number of rooms	2,670.21	587.7595	4.543032	5.23E-05	
Hours Worked, Front Desk	29.30	9.270823	3.160299	0.003044	
Hours Worked, GM	52.01	18.93043	2.747581	0.009043	
Hours Worked, Housekeeping	13.41	4.683324	2.862522	0.006728	

	2025
Gross Sales	\$ 12,165,162
Total Variable Costs	\$ 5,722,777
Contribution Margin	\$ 6,442,385
Total Fixed Costs	\$ 1,696,822
Net Revenue	\$ 4,745,563

EX 8.17B & EX 8.19

	Best-Case	Likely-Case	Worst-Case
Gross Sales	\$ 12,165,162	\$ 12,165,162	\$ 12,165,162
Total Variable Costs	\$ 6,008,916	\$ 5,837,232	\$ 6,123,371
Contribution Margin	\$ 6,156,246	\$ 6,327,930	\$ 6,041,791
Total Fixed Costs	\$ 1,696,822	\$ 1,696,822	\$ 1,696,822
Net Revenue	\$ 4,459,424	\$ 4,631,108	\$ 4,344,969



SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.6260322				
R Square	0.3919164				
Adjusted R Square	0.391612				
Standard Error	486.1264				
Observations	9994				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	5	1521272575	304254515	1287.47445	0.00
Residual	9988	2360352937	236318.876		
Total	9993	3881625512			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	221.75141	12.60104565	17.5978578	2.7291E-68	
Discount	-438.73427	24.64841598	-17.799694	8.4529E-70	
Discount \$	2.4154052	0.030141825	80.1346697	0.00	
Region West	0.1232541	15.12802917	0.0081474	0.99349955	
Region East	-17.727807	14.84929891	-1.1938481	0.23256576	
Region Central	18.590415	15.9032368	1.16897054	0.2424434	

Tình huống Ứng dụng Chuyên môn

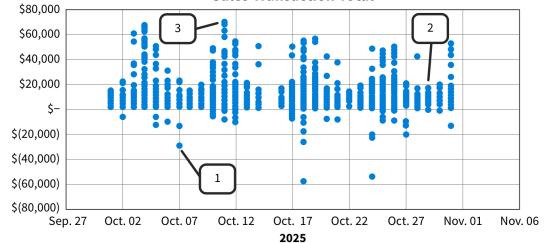
Professional Application Cases (PAC)

PAC 8.1A & PAC 8.1b

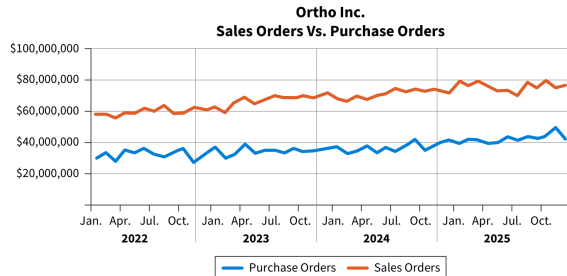
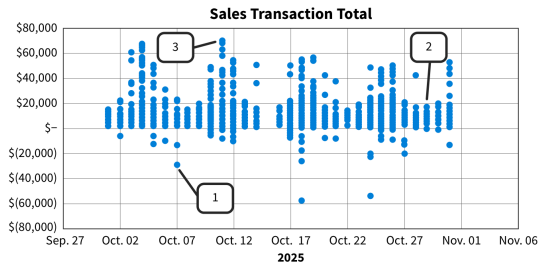
Sales Frequency Distribution

Prod ProductDescription	TR TransDate			
	Distinct Count of TR Subtotal		TR Subtotal	
	2024	2025	2024	2025
Ceramic-on-Ceramic Knee	4,378	4,196	113,844,112	116,040,741
Metal and Crosslinked Polyethylene Knee	7,217	7,208	145,197,885	153,572,138
Metal and Plastic Knee	9,567	9,714	103,546,449	107,545,175
Metal-on-Metal Knee (Cobalt chrome)	20,457	20,331	185,198,302	198,397,781
Metal-on-Metal Knee (Stainless steel)	4,153	4,306	106,320,616	116,722,920
Metal-on-Metal Knee (Titanium)	16,526	16,760	199,398,247	220,828,360
Grand Total	60,102	60,424	853,505,611	913,107,114

Sales Transaction Total



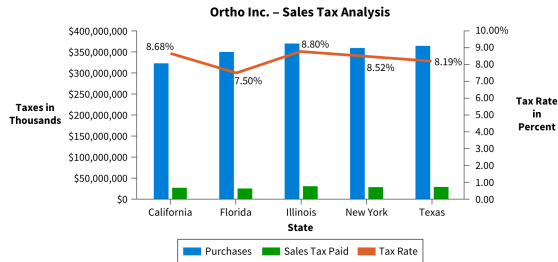
PAC 8.1A_1 & PAC 8.2



Ortho Inc. Correlation between Purchases and Sales		
	Purchase Orders	Sales Orders
Purchase Orders	1	
Sales Orders	0.750363942	1

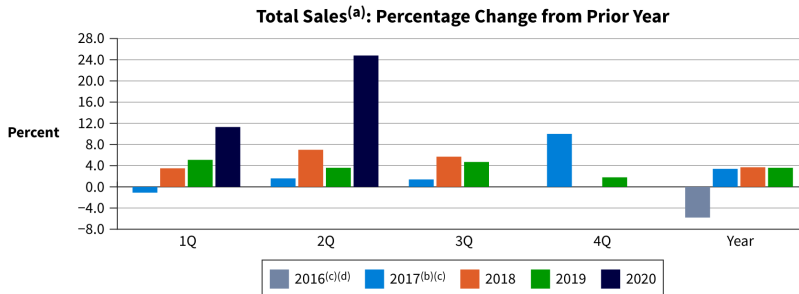
PAC 8.3 & PAC 8.4

Site	2025	2024	Difference
CA	172,934,662.11	174,888,045.18	-1,953,383.07
FL	183,493,309.15	164,530,136.81	18,963,172.34
IL	175,041,416.59	174,872,628.25	168,788.34
NY	193,364,104.19	163,660,289.73	29,703,814.46
TX	188,249,027.51	175,554,510.70	12,694,516.81
Total	913,082,519.55	853,505,610.67	59,576,908.88

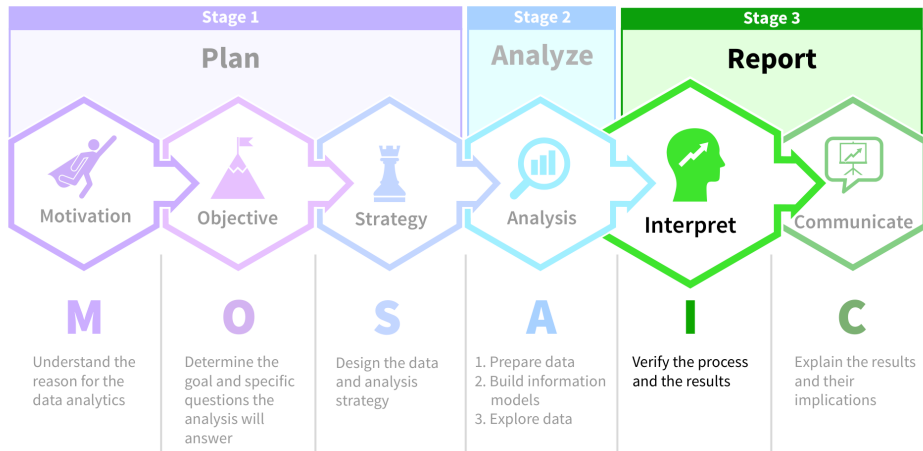


Tình huống Bỗ trợ & Dữ liệu

PR, ERD, Info



Fiscal Year	1Q	2Q	3Q	4Q	Year
2020	11.3 %	24.8 %			
2019	5.1 %	3.6 %	4.7 %	1.8 %	3.6 %
2018	3.5 %	7.0 %	5.7 %	- %	3.7 %
2017 (b) (c)	(1.1)%	1.6 %	1.4 %	10.0 %	3.4 %
2016 (c) (d)					(5.8)%



Data Dictionary

MATERIAL MASTER TABLE

Name	Primary Key	Definition
MaterialCategory	☐	Constant stating raw material
Material	☒	Material item identifier
MaterialDescription	☐	Description of material

Notes: The Material Master table contains information about different raw materials purchased by Ortho.

PRODUCT MASTER TABLE

Name	Primary Key	Definition
ProductCategory	☐	Constant stating knee implant
Product	☒	Product item identifier
ProductDescription	☐	Description of product

Notes: The Product Master table contains information about different products sold by Ortho.

TRANSACTIONS TABLE

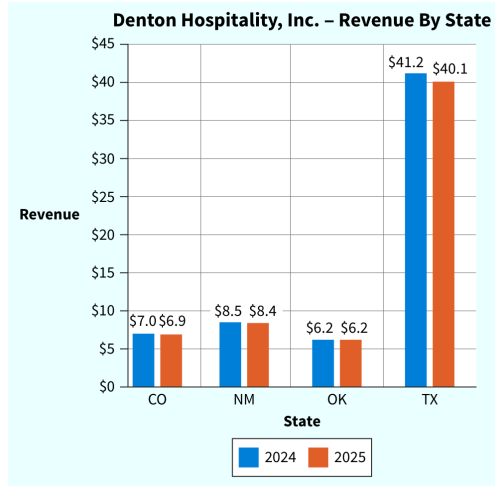
Name	Primary Key	Definition
TransType	☒	Indicates the type of transaction (O for orders and S for sales)
TransNumber	☒	Identifier for orders and sales
SiteCode	☐	Transaction site (one of five locations)
Manager	☐	Employee associated with the transaction
EntityID	☐	Supplier/Customer associated with the transaction
Quantity	☐	Transaction quantity

Apply It 8.1A

Summary of DHI Financial Performance

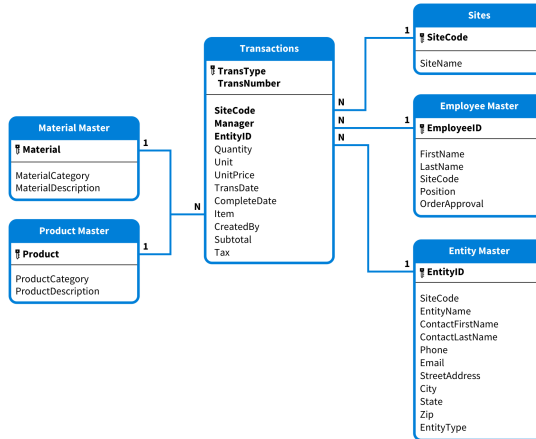
State	Number of Hotels	Revenue 2024	Revenue 2025	Profit 2024	Profit 2025
CO	5	\$ 6,972,316	\$ 6,942,074	\$ 3,701,428	\$ 3,582,689
NM	7	\$ 8,491,172	\$ 8,414,229	\$ 4,025,218	\$ 3,862,043
OK	5	\$ 6,223,433	\$ 6,156,051	\$ 3,178,742	\$ 2,928,290
TX	31	\$ 41,220,622	\$ 40,071,922	\$ 18,320,989	\$ 16,548,176
Grand Total	48	\$ 62,907,543	\$ 61,584,276	\$ 29,226,377	\$ 26,921,198

Apply It 8.1B



ERD 8.1A

Entity-Relationship Diagram (ERD)



(Notes: The ERD indicates which tables to join and which columns to use in the joins.)

Ortho Inc. Comparative Income Statement

Comparative Income Statement \$ in Millions

	2024 (audited)	2025 (unaudited)
Net Sales	\$913	\$854
Cost of Sales	<u>579</u>	<u>542</u>
Gross Profit	334	312
Research and development	57	59
Selling, general and administrative	191	196
Recall charges, net of insurance proceeds	5	5
Intangible asset amortization	<u>3</u>	<u>2</u>
Total operating expenses	<u>256</u>	<u>262</u>
Operating Income	78	50
Other income (expenses), Net	<u>1</u>	<u>2</u>
Earnings before income taxes	79	52
Income taxes	<u>11</u>	<u>8</u>
Net earnings	\$ 68	\$ 44

Unit	☐	Unit of transaction (lb or kg for orders, ea for sales)
UnitPrice	☐	Price per unit
TransDate	☐	Sales/purchase order date
CompleteDate	☐	Shipping/receiving date (sales/orders)
Item	☐	The material purchased or product sold
CreatedBy	☐	The source of the data (e.g., daily batch)
Subtotal	☐	Transaction total before tax (UnitPrice * Quantity)
Tax	☐	Tax amount

Notes: The Transactions table contains a combination of purchase order, sales order, and sales return transactions

SITES TABLE

Name	Primary Key	Definition
SiteCode	☒	Identified for company sites (five state abbreviations)
SiteName	☐	Name of site (states)

Notes: The Sites table contains the name of each site, which is the name of the state where the site is located.

EMPLOYEE MASTER TABLE

Name	Primary Key	Description
EmployeeID	☒	Employee identifier
FirstName	☐	Employee first name
LastName	☐	Employee last name
SiteCode	☐	Company site identified (state)
Position	☐	Employee position
OrderApproval	☐	Employee order approval authority

Notes: The Employee Master table contains information about employees associated with the sales order and purchase order transactions, e.g., sales and product managers.

ENTITY MASTER TABLE

Name	Primary Key	Description
EntityID	☒	Entity (supplier/customer/Ortho, Inc.) identifier
SiteCode	☐	Indicates which Ortho site the entity is associated with
EntityName	☐	Entity company name
ContactFirstName	☐	Entity contact first name
ContactLastName	☐	Entity contact last name
Phone	☐	Entity phone
Email	☐	Entity email
StreetAddress	☐	Entity street address
City	☐	Entity city
State	☐	Entity state
Zip	☐	Entity zip code
EntityType	☐	Entity type ("S" for customer and "O" for supplier)

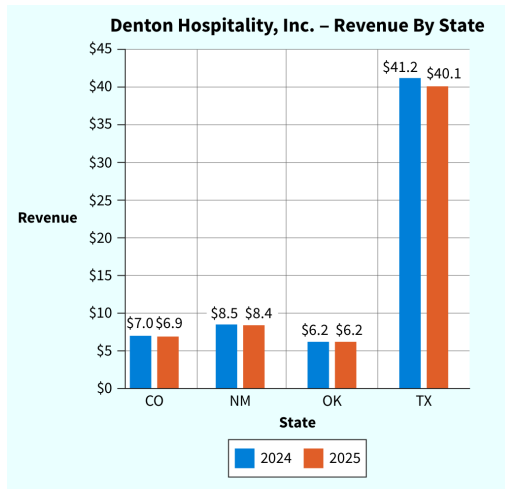
Shipping Cost Per Year

Year	Total Shipping Cost
2022	\$ 49,769.30
2023	\$ 50,198.57
2024	\$ 60,925.19
2025	\$ 77,280.73
Grand Total	\$ 238,173.79

Number of Shipments
by Shipping Mode

Priority and Shipping Mode	Number of Shipments
Critical	783
First Class	340
Same Day	154
Second Class	289
High	3,069
First Class	703
Same Day	273
Second Class	744
Standard Class	1,349
Low	432
Standard Class	432
Medium	5,710
First Class	495
Same Day	116
Second Class	912
Standard Class	4,187
Grand Total	9,994

Apply It 8.1A_1



Region	Number of Hospitals
East North Central	262
East South Central	136
Mid Atlantic	168
Mountain	160
New England	100
Pacific	259
South Atlantic	252
West North Central	214
West South Central	449
Grand Total	2,000

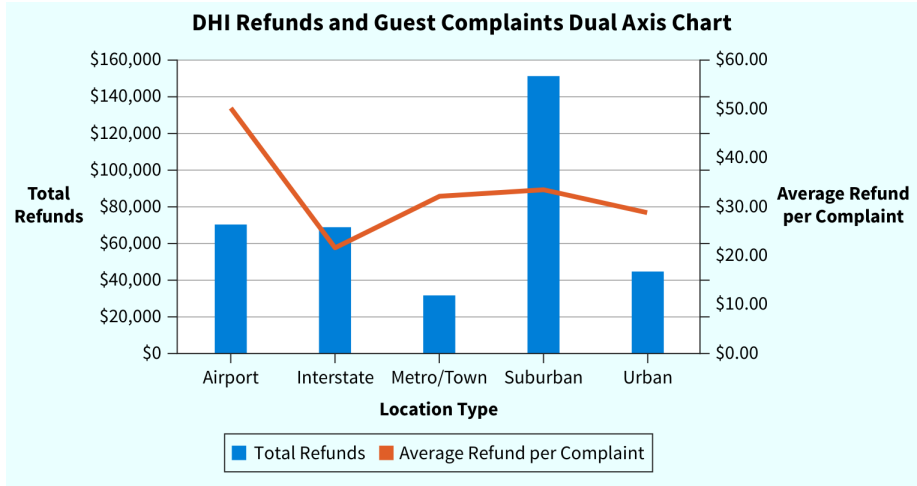
Total Expenses by Region-2025	
Region	
East North Central	\$52,941,169
East South Central	\$19,037,967
Mid Atlantic	\$60,506,505
Mountain	\$26,438,517
New England	\$25,634,270
Pacific	\$71,210,066
South Atlantic	\$57,893,598
West North Central	\$26,015,206
West South Central	\$47,082,098

Total Expense

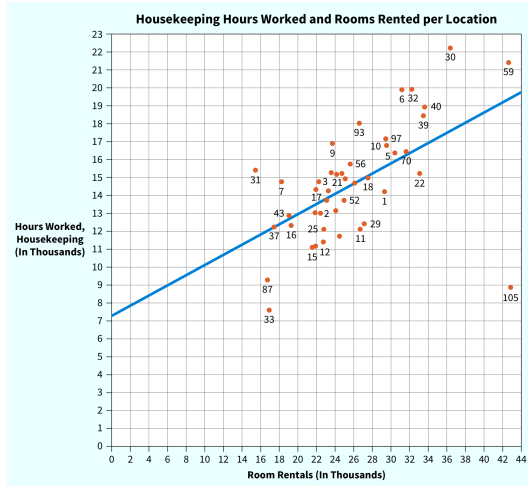


SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.982051941							
R Square	0.964426014							
Adjusted R Square	0.964301005							
Standard Error	61360.99522							
Observations	2000							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	7	2.03E+14	2.9E+13	7714.85352	0			
Residual	1992	7.5E+12	3.77E+09					
Total	1999	2.11E+14						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	-4191.07786	2275.171	-1.84209	0.06561006	-8653.042868	270.8871	-8653.04	270.8871
Admissions	6.013726131	0.387068	15.5366	1.8108E-51	5.254624717	6.772828	5.254625	6.772828
Census	-57.72122789	19.55346	-2.95197	0.00319445	-96.06861315	-19.3738	-96.0686	-19.3738
Outpatient Visits	-0.019727175	0.009985	-1.97573	0.04832264	-0.039308781	-0.00015	-0.03931	-0.00015
Births	-9.977260409	1.656594	-6.02276	2.0367E-09	-13.22609855	-6.72842	-13.2261	-6.72842
Payroll Expense	1.804388653	0.036122	49.95226	0.00000000	1.733547274	1.87523	1.733547	1.87523
Personnel	30.9359429	2.697418	11.46872	1.5657E-29	25.64588566	36.226	25.64589	36.226
Births or Not	-5932.874462	3329.682	-1.78181	0.07493183	-12462.8985	597.1496	-12462.9	597.1496

Apply It 8.3



Apply It 8.4



Apply It 8.5

Regression Statistics					
Multiple R	0.7215151				
R Square	0.520584				
Adjusted R Square	0.4714132				
Standard Error	93901.466				
Observations	44				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	3.73412E+11	9.3353E+10	10.58724498	6.62556E-06
Residual	39	3.43882E+11	8817485253		
Total	43	7.17293E+11			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	413314.47	107148.5543	3.85739665	0.000418405	
Room Rentals	11.015461	2.918679199	3.77412543	0.000534577	
Hours Worked, Housekeeping	6.0934507	5.531393108	1.1016123	0.277382666	
Age	-3565.695	1674.938087	-2.12885155	0.039637083	
Employees, Maintenance	16762.392	6906.902129	2.42690446	0.019949355	

Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!